CSA4707-DEEP LEARNING

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Experiment 3: Probability Distributions and Bayes' Theorem

Problem Statement: A diagnostic centre uses a screening test for a disease. Doctors need the probability that a patient with a positive result actually has the disease. Disease prevalence is 1%, test sensitivity 95% and specificity 90%. Patient age and test-score data follow known distributions. The objective is to model data with standard distributions and apply Bayes' theorem to a real decision. Use NumPy, SciPy, Matplotlib.

Tasks:

1. Generate and plot Bernoulli, Binomial, Poisson, Uniform and Gaussian samples with SciPy; compare with their PDFs/PMFs.
2. Compute P(Disease | Positive) using Bayes' theorem.
3. Simulate 100,000 patients and verify the result empirically.
4. Plot how the posterior changes as prevalence varies from 0.1% to 20%.

Architecture: Probability Distributions and Bayes' Theorem

Start

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Import NumPy, SciPy and Matplotlib

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Generate Random Samples

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Bernoulli, Binomial, Poisson, Uniform and Gaussian Distributions

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Plot Distributions and Compare Results

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Apply Bayes' Theorem

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Calculate Disease Probability After a Positive Test

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Simulate 100,000 Patients

↓ Verify the Probability

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Vary Disease Prevalence

↓

Plot Prevalence vs Posterior Probability

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End

PROGRAM CODE:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import bernoulli, binom, poisson, uniform, norm

np.random.seed(10)

data = [
    ("Bernoulli", bernoulli(0.5), bernoulli.rvs(0.5, size=1000)),
    ("Binomial", binom(10, 0.5), binom.rvs(10, 0.5, size=1000)),
    ("Poisson", poisson(4), poisson.rvs(4, size=1000)),
    ("Uniform", uniform(), uniform.rvs(size=1000)),
    ("Gaussian", norm(), norm.rvs(size=1000))
]

plt.figure(figsize=(10, 7))

for i, (name, dist, sample) in enumerate(data):
    plt.subplot(3, 2, i + 1)
    plt.hist(sample, bins=20, density=True, alpha=0.5)

    if i < 3:
        x = np.arange(int(min(sample)), int(max(sample)) + 1)
        plt.plot(x, dist.pmf(x), "ro-")
    else:
        x = np.linspace(min(sample), max(sample), 100)
        plt.plot(x, dist.pdf(x), "r-")

    plt.title(name)

plt.tight_layout()
```

```
plt.show()

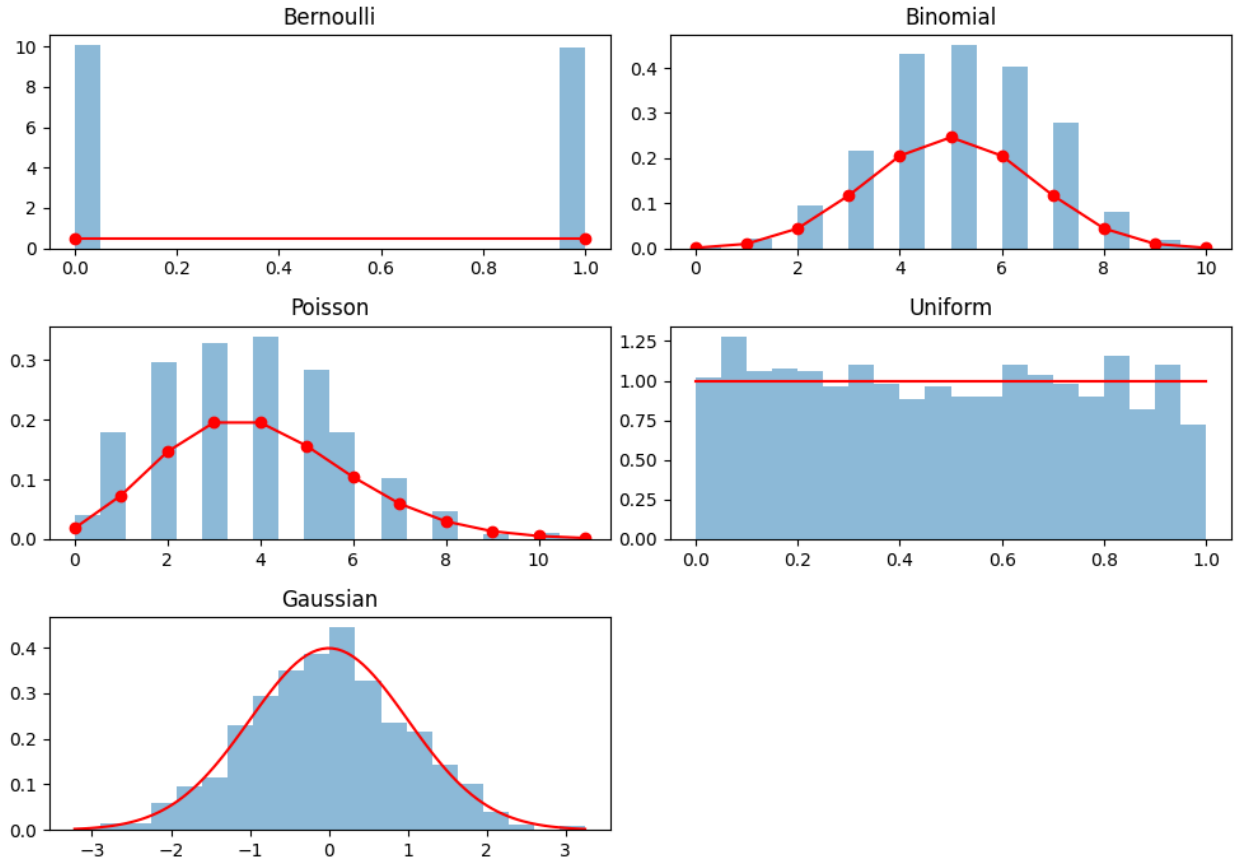
p, s, c = 0.01, 0.95, 0.90
posterior = (s * p) / (s * p + (1 - c) * (1 - p))
print("Bayes Probability:", round(posterior * 100, 2), "%")

n = 100000
disease = np.random.random(n) < p
positive = np.where(disease,
                    np.random.random(n) < s,
                    np.random.random(n) < (1 - c))
print("Simulated Probability:",
      round(np.mean(disease[positive]) * 100, 2), "%")

p = np.linspace(0.001, 0.20, 200)
post = s * p / (s * p + (1 - c) * (1 - p))

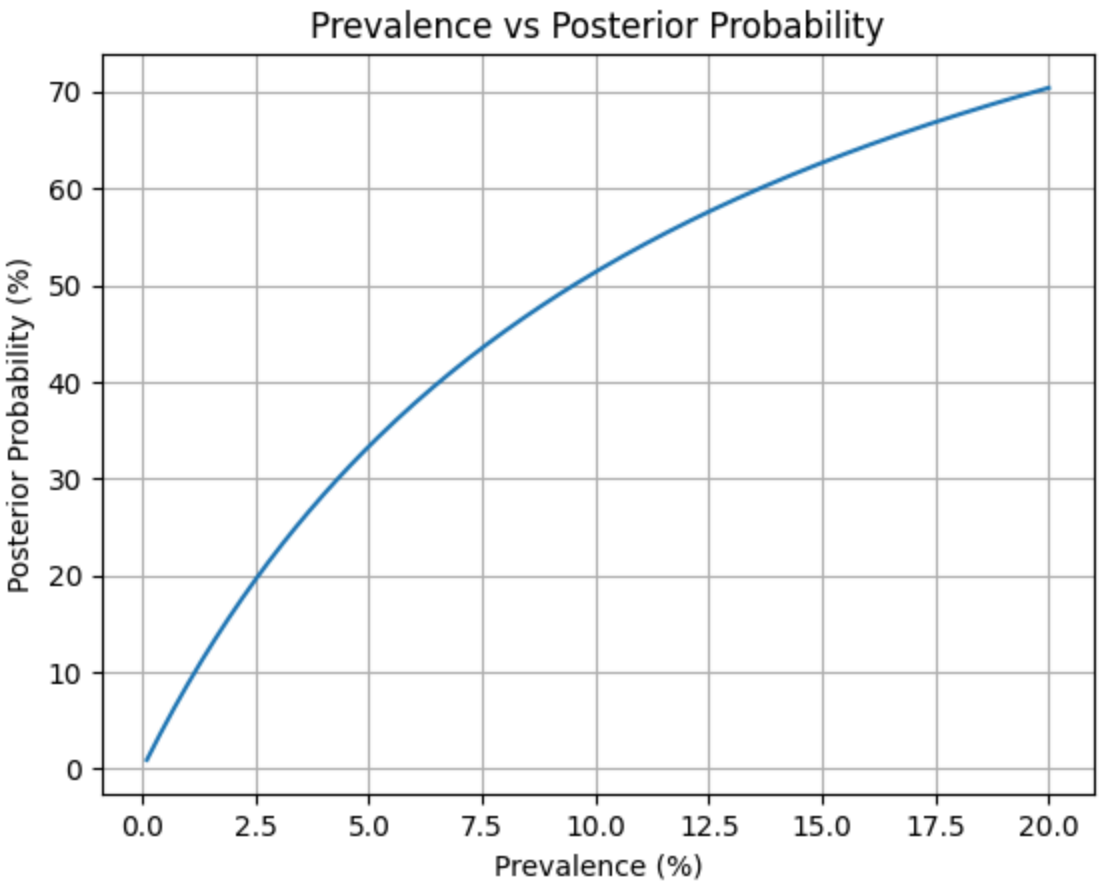
plt.plot(p * 100, post * 100)
plt.xlabel("Prevalence (%)")
plt.ylabel("Posterior Probability (%)")
plt.title("Prevalence vs Posterior Probability")
plt.grid()
plt.show()
```

OUTPUT:



Bayes Probability: 8.76 %

Simulated Probability: 8.99 %


Prevalence vs Posterior Probability
Posterior Probability (%)
70
60
50
40
30
20
10
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Prevalence (%)